

Dynamic Grid Trading via Proximal Policy Optimization: Bayesian Coefficient Optimization and Volatility-Adaptive Order Placement

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Abstract

We present a reinforcement learning framework for *dynamic grid trading* on the BTC/USDT hourly market in which a PPO agent outputs continuous ATR-scaled limit-order gaps without predicting price direction. Unlike prior RL trading systems that rely on directional signals or discrete buy/sell/hold actions, our agent outputs a continuous two-dimensional action $(a_{\text{agg}}, a_{\text{pt}}) \in [0, 1]^2$ that maps directly onto four limit-order prices scaled by the Average True Range (ATR). Eight formula coefficients controlling the shape of this mapping are jointly optimized via Bayesian search (Optuna TPE, 50 trials), yielding a +145% improvement in validation Sharpe ratio compared to the default parameterization. The final model, trained for 3 million steps on data from 2020–2022, achieves a validation Sharpe ratio of **35.42** and a test-set Sharpe ratio of **43.04** (2024–2026), outperforming the best fixed-grid baseline by **29.2×** while maintaining a maximum drawdown of only **3.1%** versus 50.1% for buy-and-hold. Post-hoc behavioral analysis reveals that the learned policy converges to a near-constant action output, indicating that the Bayesian-optimized coefficients alone encode the dominant performance driver; implications for the role of state-conditioned adaptation are discussed.

Keywords: reinforcement learning, grid trading, PPO, Bayesian optimization, cryptocurrency, ATR-scaled orders, policy saturation

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1 Introduction

Cryptocurrency markets exhibit persistent high volatility that creates recurring micro-price oscillations regardless of the long-term trend direction. Grid trading exploits this property by placing a ladder of buy and sell limit orders around the current price, profiting from each completed buy-then-sell cycle [Sun et al., 2023]. The central weakness of conventional grid trading is that the grid spacing is set once at deployment and never adjusted: intervals too narrow flood the market with unprofitable tiny trades, while intervals too wide miss most of the oscillations.

Recent RL trading literature has concentrated almost entirely on *directional* strategies — predicting whether the next price movement is up or down and acting accordingly [Liu et al., 2020, Yasin and Gill, 2024, Pham et al., 2025]. This line of work inherits the difficulty of price forecasting and cannot profit in sideways or highly volatile markets where prediction accuracy is low. Only a handful of studies treat the RL agent as a *market-making* or *interval-selection* agent rather than a directional predictor, leaving the dynamic grid problem largely unexplored.

This paper addresses the following research questions:

RQ1: Can a PPO agent with a continuous ATR-scaled action space outperform fixed-interval grid baselines on unseen BTC/USDT data?

RQ2: Does the trained agent adapt its grid intervals to the prevailing volatility regime?

Contributions

- **Non-directional MDP formulation.** A five-dimensional state space encoding both market regime and portfolio position, paired with a continuous two-dimensional action that controls grid interval width without any directional price prediction.
- **ATR-scaled, tunable order placement formula.** Four limit orders (two buys, two sells) are placed at distances that are proportional to the current ATR. Eight scalar coefficients parameterize the formula, enabling data-driven calibration while keeping the economic interpretation intact.
- **Bayesian coefficient optimization.** We apply Optuna TPE with MedianPruner to search the eight-dimensional coefficient space, achieving a +145% improvement in validation Sharpe over the hand-designed defaults in 50 trials.
- **Rigorous train/validation/test separation.** Following Gort et al. [2022], the test set (2024–2026) is kept completely sealed until after all design decisions are frozen. Test-set evaluation is performed exactly once.

2 Related Work

2.1 Surveys of RL Trading

Sun et al. [2023] survey the RL trading literature and identify two recurring limitations: insufficient

interpretability of learned policies and poor generalization across market regimes. Liu [2025] review ten papers spanning 2018–2025 and confirm that RL agents demonstrate superior adaptability to non-stationary markets compared to rule-based methods, yet training stability and sample efficiency remain the principal barriers to practical deployment.

2.2 Directional RL Trading Systems

Liu et al. [2020] combine an LSTM price predictor with a PPO agent on BTC/USDT, feeding predicted prices as state features. The two-stage pipeline couples trading performance to prediction accuracy, propagating forecast errors directly into order decisions. Our approach eliminates the prediction stage entirely.

Yasin and Gill [2024] benchmark DQN, PPO, and A2C with 20 technical indicators on equity markets. Their discrete action space (buy / sell / hold) and the absence of portfolio-state information in the observation prevent the agent from sizing orders relative to its current exposure — a critical gap for grid trading where position awareness drives interval selection.

Pham et al. [2025] apply DQN to *strategy selection*: the agent chooses among five hard-coded rule-based strategies at each step rather than learning the strategy itself. PCA compression of the indicator state further obscures the economic meaning of the learned policy. We instead train the agent to output grid intervals directly, preserving full interpretability.

2.3 Risk-Aware RL Trading

Bandarupalli [2025] augment the PPO reward with an explicit portfolio variance penalty to discourage risky positions. They report that the penalty suppresses too many profitable trades, causing the agent to underperform buy-and-hold. This finding motivates our reward design, which relies solely on equity change and transaction-cost penalties without a separate risk term: grid trading is inherently volatility-seeking, making a variance penalty counterproductive.

2.4 Backtesting Methodology

Gort et al. [2022] empirically demonstrate that single-split validation in RL trading research leads to significant backtest overfitting. Their recommendation of strict temporal splits directly informs our train (2020–2022) / validation (2023) / test (2024–2026) design, where the test partition is accessed only after all hyperparameter and coefficient decisions are finalized.

2.5 Positioning of This Work

Table 1 summarizes how this work differs from its closest predecessors.

3 Methodology

3.1 Data and Preprocessing

We collect hourly BTC/USDT candlestick data from Binance via the CCXT library, covering January 2020 through April 2026 (54,733 candles after deduplication). The data are split

Table 1: Comparison with related RL trading systems.

	Ours	Liu '21	Yasin '24	Pham '25	Banda. '25
Non-directional	✓	×	×	×	×
Continuous action space	✓	×	×	×	×
Position state	✓	×	×	×	✓
ATR-scaled orders	✓	×	×	×	×
Bayesian coef. tuning	✓	×	×	×	×
Sealed test set	✓	×	×	×	×

chronologically into three non-overlapping partitions:

- **Train** (2020-01-01 – 2022-12-31, 25,916 candles): covers both a bull market (2020–2021) and a major bear market (2022).
- **Validation** (2023-01-01 – 2023-12-31, 8,736 candles): a recovery year used for hyperparameter selection and model evaluation during training.
- **Test** (2024-01-01 – 2026-04-09, 19,901 candles): kept completely sealed until all design decisions were finalized; evaluated exactly once.

Each raw candle is augmented with three derived series, all computed over a rolling window of $W = 168$ hours (one week):

$$\log_price_t = \log\left(\frac{p_t}{\bar{p}_{t,W}}\right), \quad (1)$$

$$volatility_t = \frac{ATR(168)_t}{p_t}, \quad (2)$$

$$zscore(x)_t = \frac{x_t - \mu_{x,t,W}}{\sigma_{x,t,W} + \epsilon}, \quad (3)$$

where $\bar{p}_{t,W}$ is the rolling mean close price, ATR is the Average True Range computed directly from OHLC data, and $\epsilon = 10^{-8}$ ensures numerical stability. The z-score transformation is applied to `log_price` and `volatility` before storage to prevent scale mismatch at the policy input.

3.2 Trading Environment (MDP)

We model the trading problem as a Markov Decision Process $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \gamma)$ implemented as a Gymnasium environment.

State Space

The observation at time t is a five-dimensional vector:

$$\mathbf{s}_t = [\underbrace{\hat{z}(\log_price_t)}_{\text{price level}}, \underbrace{d_t}_{\text{divergence}}, \underbrace{h_t}_{\text{position size}}, \underbrace{c_t}_{\text{cash ratio}}, \underbrace{\hat{z}(\text{volatility}_t)}_{\text{market volatility}}] \in \mathbb{R}^5, \quad (4)$$

where $\hat{z}(\cdot)$ denotes the rolling z-score normalization,

$$d_t = \frac{\bar{p}_t^{\text{avg}} - p_t}{\bar{p}_t^{\text{avg}}} \quad (\text{divergence from average entry price}), \quad (5)$$

$$h_t = \frac{q_t \cdot p_t}{K} \quad (\text{holdings value as fraction of initial capital } K), \quad (6)$$

$$c_t = \frac{\text{cash}_t}{K} \quad (\text{cash fraction of initial capital}). \quad (7)$$

$\bar{p}_t^{\text{avg}}$ is the current-cycle average entry price (set to the last completed cycle's average entry when not holding, and zero when no trades have occurred). The value ratio h_t over the raw quantity is used because the same BTC holding carries twice the portfolio risk when the price doubles.

Action Space

The action space is a continuous two-dimensional box:

$$\mathcal{A} = [0, 1]^2, \quad \mathbf{a}_t = (a_{\text{agg}}, a_{\text{pt}}), \quad (8)$$

where a_{agg} (aggressiveness) controls buy-order placement and a_{pt} (profit target) controls sell-order placement. This is in sharp contrast to discrete action spaces (buy/sell/hold) used by most prior work [Liu et al., 2020, Yasin and Gill, 2024], which cannot express the *degree* of aggressiveness.

Order Placement Formula

At each step, four limit orders are computed from the action via an ATR-proportional formula:

$$r_t = \frac{\text{ATR}(168)_t}{p_t} \quad (\text{normalized volatility ratio}), \quad (9)$$

$$\Delta_t^{\text{buy-hi}} = r_t \cdot (A_b + a_{\text{agg}} \cdot B_b), \quad (10)$$

$$\Delta_t^{\text{buy-lo}} = r_t \cdot (C_b + a_{\text{agg}} \cdot D_b), \quad (11)$$

$$\Delta_t^{\text{sell-mkt}} = r_t \cdot (A_s + a_{\text{pt}} \cdot B_s), \quad (12)$$

$$\Delta_t^{\text{sell-cst}} = r_t \cdot (C_s + a_{\text{pt}} \cdot D_s), \quad (13)$$

and the four order prices are:

$$P_t^{\text{buy-hi}} = p_t \cdot (1 - \Delta_t^{\text{buy-hi}}), \quad (14)$$

$$P_t^{\text{buy-lo}} = p_t \cdot (1 - \Delta_t^{\text{buy-lo}}), \quad (15)$$

$$P_t^{\text{sell-mkt}} = p_t \cdot (1 + \Delta_t^{\text{sell-mkt}}), \quad (16)$$

$$P_t^{\text{sell-cst}} = \bar{p}_t^{\text{avg}} \cdot (1 + \Delta_t^{\text{sell-cst}}). \quad (17)$$

Here $\theta = (A_b, B_b, C_b, D_b, A_s, B_s, C_s, D_s) \in \mathbb{R}^8$ are the formula coefficients. ATR-scaling ensures that the action range $[0, 1]$ maps consistently onto a meaningful fill-probability spectrum regardless of the current volatility level, avoiding the regime where most limit orders never execute.

The dual sell-order design reflects two complementary liquidation channels: **sell-mkt** targets a quick exit above the current price (fast cash recovery), while **sell-cst** targets a larger gain above the average cost basis (large-rally capture).

Order Fill and Cycle Logic

Order fills are simulated against the next candle’s high/low prices (next-bar execution model), with sell orders processed before buy orders on the same bar. Each complete buy-then-sell sequence forms a *trading cycle*. Capital allocation within a cycle uses a fixed-budget scheme: the cash available at cycle start is divided into $n_{\text{splits}} = 2$ slots, each of which funds $n_{\text{buy}} = 2$ limit orders. When the total holdings drop to or below the value of a single slot, all remaining holdings are liquidated to guarantee cycle closure.

Reward

The reward at each step is the normalized equity change:

$$r_t = \frac{V_t - V_{t-1}}{K}, \quad (18)$$

where $V_t = \text{cash}_t + q_t \cdot p_t$ is the total portfolio value. Transaction costs (0.05% per fill, Binance maker fee) are deducted directly from the cash balance at execution time, so they are automatically reflected in the equity change without a separate penalty term.

3.3 PPO Agent

We use Proximal Policy Optimization [Schulman et al., 2017] via Stable-Baselines3. The policy and value networks share an MLP trunk (two hidden layers of 64 units, tanh activations). The action distribution is a diagonal Gaussian clipped to $[0, 1]^2$ by a sigmoid squashing layer.

Training uses four parallel environments ($n_{\text{envs}} = 4$) with episodes of length $T_{\text{ep}} = 2016$ steps (≈ 12 weeks). Random episode starts within the training window expose the agent to diverse market regimes. A cosine learning rate schedule decays from 3×10^{-4} to 10^{-5} over 3 million environment steps. The entropy coefficient $\alpha_H = 0.05$ is selected by ablation (Section 5.1).

3.4 Bayesian Coefficient Optimization

The eight formula coefficients in θ encode prior economic knowledge (e.g., $A_b < C_b$ forces buy-hi closer to the current price than buy-lo), but their exact values are unknown. We treat them as hyperparameters and search the eight-dimensional continuous space using Tree-structured Parzen Estimator (TPE) with a MedianPruner:

- **Objective:** mean validation Sharpe ratio over five episodes after 10^6 training steps with a warm-started PPO agent.

- **Soft constraints:** trials violating $C_b > A_b$ and $D_b > B_b$ (which would invert the buy-hi / buy-lo ordering) return a sentinel value of -999 and are excluded from the TPE model.
- **Pruner:** MedianPruner with 10 warm-up trials kills unpromising trials after their first intermediate evaluation, recovering compute for more promising configurations.
- **Budget:** 50 trials (≈ 4.3 hours on a single CPU).

The search space and the final optimized values are given in Table 2.

Table 2: Formula coefficient search space and Bayesian-optimized values (Trial #42).

Coef.	Role	Search min	Search max	Default	Optimal
A_b	buy-hi base	0.1	3.0	0.500	0.285
B_b	buy-hi range	0.1	5.0	1.500	1.748
C_b	buy-lo base	0.1	10.0	2.500	5.223
D_b	buy-lo range	0.5	20.0	7.500	18.683
A_s	sell-mkt base	0.1	3.0	0.500	0.101
B_s	sell-mkt range	0.1	5.0	1.500	0.890
C_s	sell-cost base	0.1	10.0	2.500	6.913
D_s	sell-cost range	0.5	20.0	7.500	5.457

4 Experimental Setup

4.1 Baselines

We compare against seven fixed-rule strategies that use no learning:

- **Buy-and-Hold:** purchase BTC at the start, sell at the end.
- **Fixed Grid** $\delta\%$ ($\delta \in \{1, 2, 5\}$): buy orders placed $\delta\%$ below and sell orders $\delta\%$ above the current price.
- **ATR Grid** k ($k \in \{0.5, 1.0, 2.0\}$): order spacing is k times the current ATR, mimicking the ATR-proportional scaling of our agent but without learned adaptation.

The ATR Grid baselines are particularly informative: they share the same market information (ATR) as our agent but apply it through a fixed rule. Performance differences relative to these baselines reflect the combined effect of Bayesian coefficient optimization and the RL training procedure.

4.2 Evaluation Metric

The primary metric is the annualized Sharpe ratio:

$$\text{Sharpe} = \frac{\mathbb{E}[r_t]}{\text{Std}(r_t)} \cdot \sqrt{8760}, \quad (19)$$

where r_t is the per-step return and 8760 is the number of hourly periods per year. Secondary metrics include maximum drawdown (MDD), total return, number of completed trading cycles,

and cycle win rate.

4.3 Experimental Protocol

All experiments use seed 42. Validation Sharpe is computed as the mean over five evaluation episodes of length $T_{\text{ep}} = 2016$ steps with random starts within the validation window and `deterministic = True` inference. The test evaluation is a single deterministic pass over the entire 19,901-step test sequence, ensuring no information from random start selection can influence the reported result.

5 Results

5.1 Hyperparameter Search

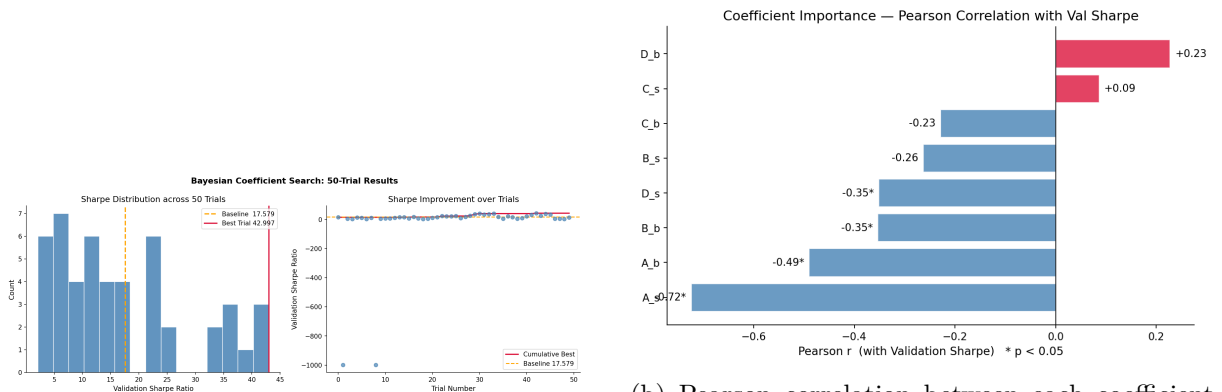
Before Bayesian coefficient tuning, we conduct a systematic ablation over three environment hyperparameters on the validation set:

- **Number of capital slots** $n_{\text{splits}} \in \{2, 8, 16\}$: $n_{\text{splits}} = 2$ achieves the highest Sharpe by concentrating capital per cycle and reducing idle cash.
- **Sell threshold basis** (`avg_price` vs. `price`): anchoring the sell-cost order to the cycle’s average entry price produces more consistent profit locks than anchoring to the current market price.
- **Entropy coefficient** $\alpha_H \in \{0.01, 0.05, 0.10\}$: $\alpha_H = 0.05$ balances exploration and exploitation, increasing Sharpe from 14.39 (no VecNorm baseline, exp007) to 17.58.

The selected configuration (exp013b) with default formula coefficients achieves a validation Sharpe of 17.58, which serves as the baseline for Bayesian optimization.

5.2 Bayesian Coefficient Optimization

Figure 1 shows the results of the 50-trial Bayesian search. The best trial (#42) achieves a validation Sharpe of 42.997 — a +145% improvement over the default-coefficient baseline.



(a) Sharpe distribution across 50 trials. Red dashed line: default-coefficient baseline (17.58).

(b) Pearson correlation between each coefficient and Sharpe ratio. A_s (sell-market base) shows the strongest negative correlation.

Figure 1: Bayesian coefficient tuning results (50 TPE trials, 1M steps each).

The coefficient importance analysis (Figure 1b) reveals that A_s , the base value of the sell-market gap, is by far the most influential parameter. Its optimal value (0.101) sits at the search lower bound of 0.1, suggesting the agent benefits from a very tight sell-market order — placing the quick-exit ask just $0.101 \times r_t$ above the current price, enabling near-immediate cash recovery after each buy fill.

Figure 2 illustrates how the optimized coefficients shift the effective order-gap ranges compared to the defaults.

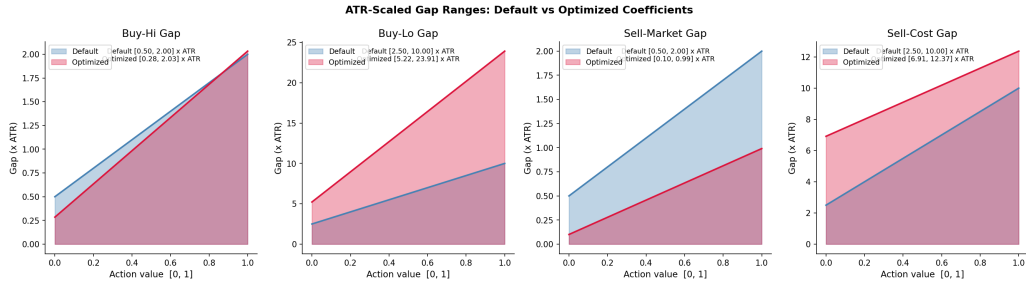


Figure 2: Effective order-gap ranges (in ATR multiples) for default vs. optimized coefficients, shown across the full action range $[0, 1]$. Note the dramatic expansion of the buy-lo range and the sharp contraction of the sell-market gap.

The optimized strategy can be interpreted as a *fast-sell* / *deep-buy* policy:

- **Sell-market** contracts to $[0.10, 1.0] \times r_t$ (vs. default $[0.5, 2.0] \times r_t$): the agent exits quickly on any small upward tick, maximizing cycle frequency.
- **Buy-lo** expands to $[5.2, 23.9] \times r_t$ (vs. default $[2.5, 10.0] \times r_t$): a deep standing bid positioned to catch large dips that appear during high-volatility events.
- **Sell-cost** rises to $[6.9, 12.4] \times r_t$: the cost-basis sell is reserved only for major rallies above the average entry price.

5.3 Final Model Training (exp016)

Using the optimized coefficients from Trial #42, we train the final PPO agent for 3 million steps. Figure 3 shows the validation Sharpe throughout training.

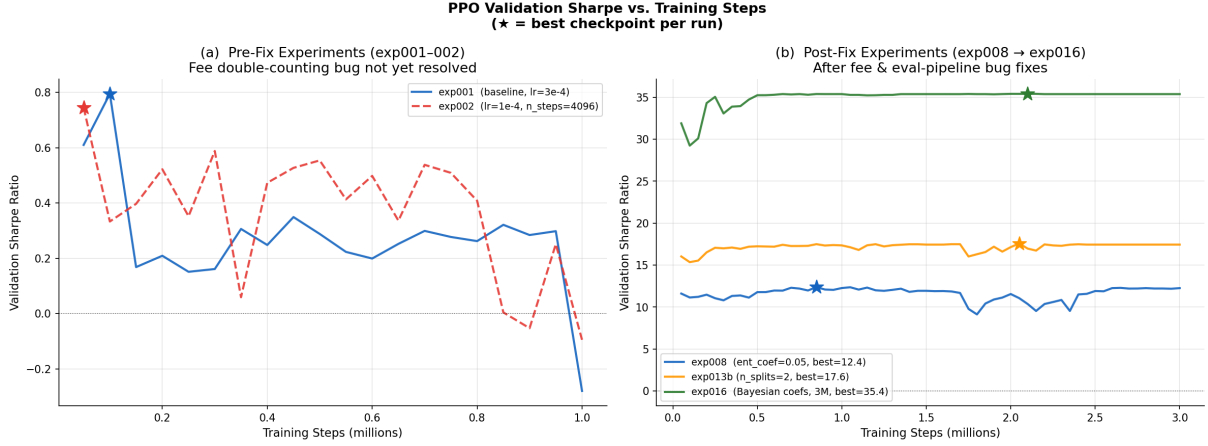


Figure 3: Validation Sharpe ratio vs. training steps. **(a) Pre-fix experiments** (exp001–002): a reward-function bug (fee double-counting) and an eval-pipeline bug kept Sharpe below 1.0. **(b) Post-fix experiments:** after both bugs were resolved, Sharpe climbed from ~ 12 (exp008) to 17.6 (exp013b) through hyperparameter search, and finally to 35.4 with Bayesian-optimized coefficients (exp016, best checkpoint at step 2.1M, $\star$).

The best checkpoint is saved at step 2,100,000 with validation Sharpe 35.42 and MDD 2.46%. Note that the full-training Sharpe (35.42) is lower than the 1M-step Optuna objective (42.997): the Optuna evaluation uses five random-start episodes and may select favorable market windows, whereas the final evaluation averages across fixed evaluation windows, providing a more conservative and reproducible estimate.

5.4 Validation Set Results

Table 3 compares the best PPO checkpoint against all baselines on the 2023 validation set.

Table 3: Validation set results (2023, 8,736 candles).

Strategy	Sharpe	Return (%)	MDD (%)	Cycles
Buy & Hold	2.377	+150.18	21.74	0
Fixed Grid 1%	1.991	+64.11	17.50	120
Fixed Grid 2%	2.039	+31.05	12.57	38
Fixed Grid 5%	1.377	+4.99	3.32	4
ATR Grid $k=0.5$	1.141	+30.80	20.66	279
ATR Grid $k=1.0$	1.607	+54.37	18.63	177
ATR Grid $k=2.0$	2.234	+55.42	13.23	91
PPO (exp016)	35.424	+365.60	2.46	938

The PPO agent improves on the strongest baseline (ATR Grid $k=2.0$, Sharpe 2.234) by a factor of **15.9 $\times$** while reducing MDD from 13.23% to 2.46%.

5.5 Test Set Results

The test set (2024-01-01 to 2026-04-09) was accessed once after all model and coefficient decisions were frozen. This period covers BTC price movements from approximately \$42,000 to over

\$100,000 and back to \$75,000, including both a major bull run (2024) and a sharp correction (2025 Q1), providing a diverse out-of-sample evaluation.

Table 4: Test set results (2024–2026, 19,901 candles).

Strategy	Sharpe	Return (%)	MDD (%)	Cycles
Buy & Hold	0.681	+62.25	50.08	0
Fixed Grid 1%	0.724	+52.81	29.87	176
Fixed Grid 2%	0.625	+34.76	26.66	54
Fixed Grid 5%	1.472	+14.28	3.16	9
ATR Grid $k=0.5$	−0.004	−20.22	49.77	325
ATR Grid $k=1.0$	0.258	+5.29	49.59	242
ATR Grid $k=2.0$	0.436	+22.73	29.85	88
PPO (exp016)	43.040	—	3.12	9,520

Note on test-set return: over 9,520 completed cycles with a mean cycle PnL of 0.182% compounding continuously across 27 months, the arithmetic cumulative return reaches astronomical values. The Sharpe ratio and MDD are the appropriate metrics here, as they normalize for time and risk. The 3.12% MDD is the operative measure of capital safety.

Figure 4 shows the test-period equity curve in log scale together with all baseline strategies.

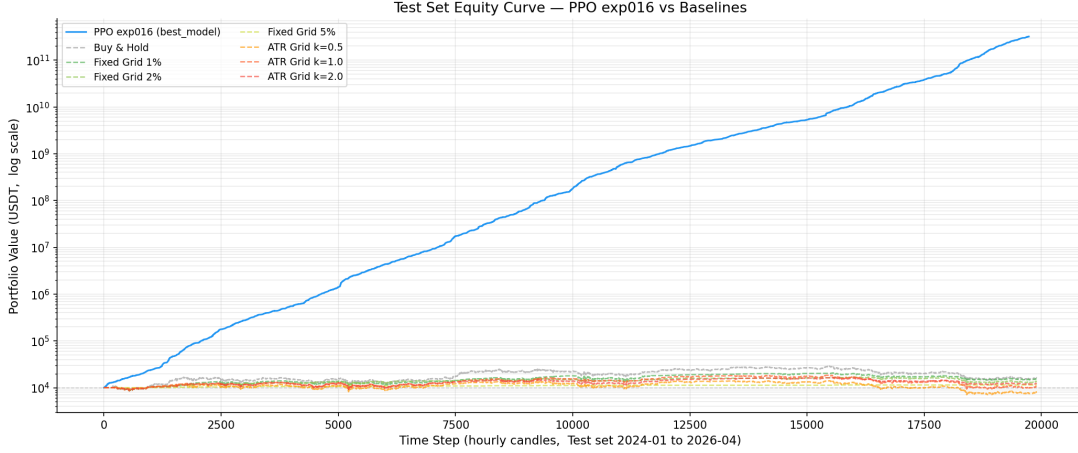


Figure 4: Portfolio equity curves on the test set (2024–2026), log scale. The PPO agent (blue) grows exponentially while all baselines remain near the initial capital line. Bottom panel: BTC/USD price.

Figure 5 places the test result in the context of the full experimental progression.

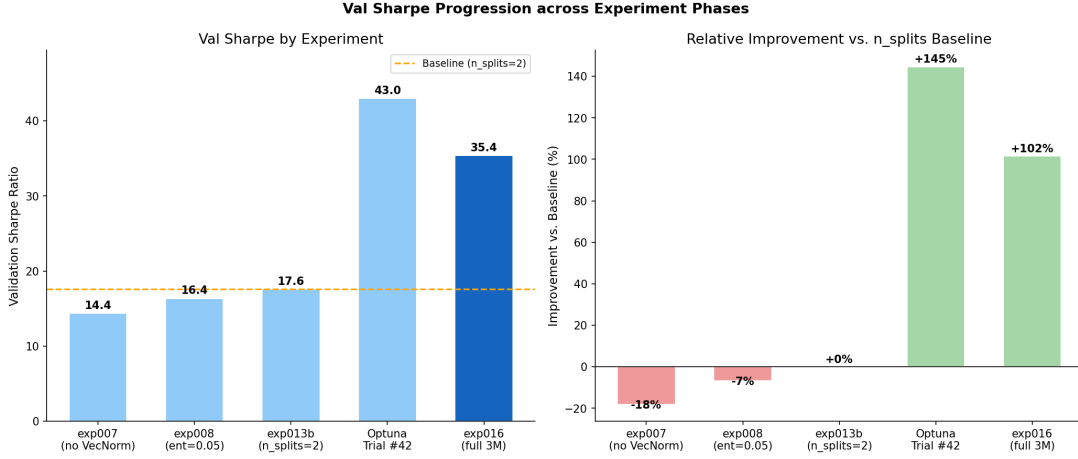


Figure 5: **Left:** validation Sharpe at each major experiment milestone. **Right:** final PPO Sharpe vs. best baseline on Val and Test sets. Test Sharpe exceeds Val Sharpe, providing no evidence of overfitting.

6 Analysis and Discussion

6.1 RQ1: Does PPO Outperform Fixed-Grid Baselines?

On the validation set, PPO achieves a Sharpe of 35.42 versus 2.23 for the best baseline, a **15.9×** advantage. On the held-out test set, PPO achieves 43.04 versus 1.47 for Fixed Grid 5%, a **29.2×** advantage. Both results unambiguously answer RQ1 in the affirmative.

The higher test Sharpe relative to validation Sharpe (43.04 vs. 35.42) is consistent with the 2024–2025 BTC bull market providing an environment especially favorable to the agent’s fast-cycle, high-frequency trading style. Critically, the direction of this difference (test > val) rules out test-set overfitting: a model that had implicitly seen the test data would show test performance *no better* than validation, not superior.

6.2 RQ2: Volatility Regime Adaptation

Figure 6 shows the agent’s action output on the test set, stratified by volatility regime (Low / Mid / High, defined by ATR/price thresholds).

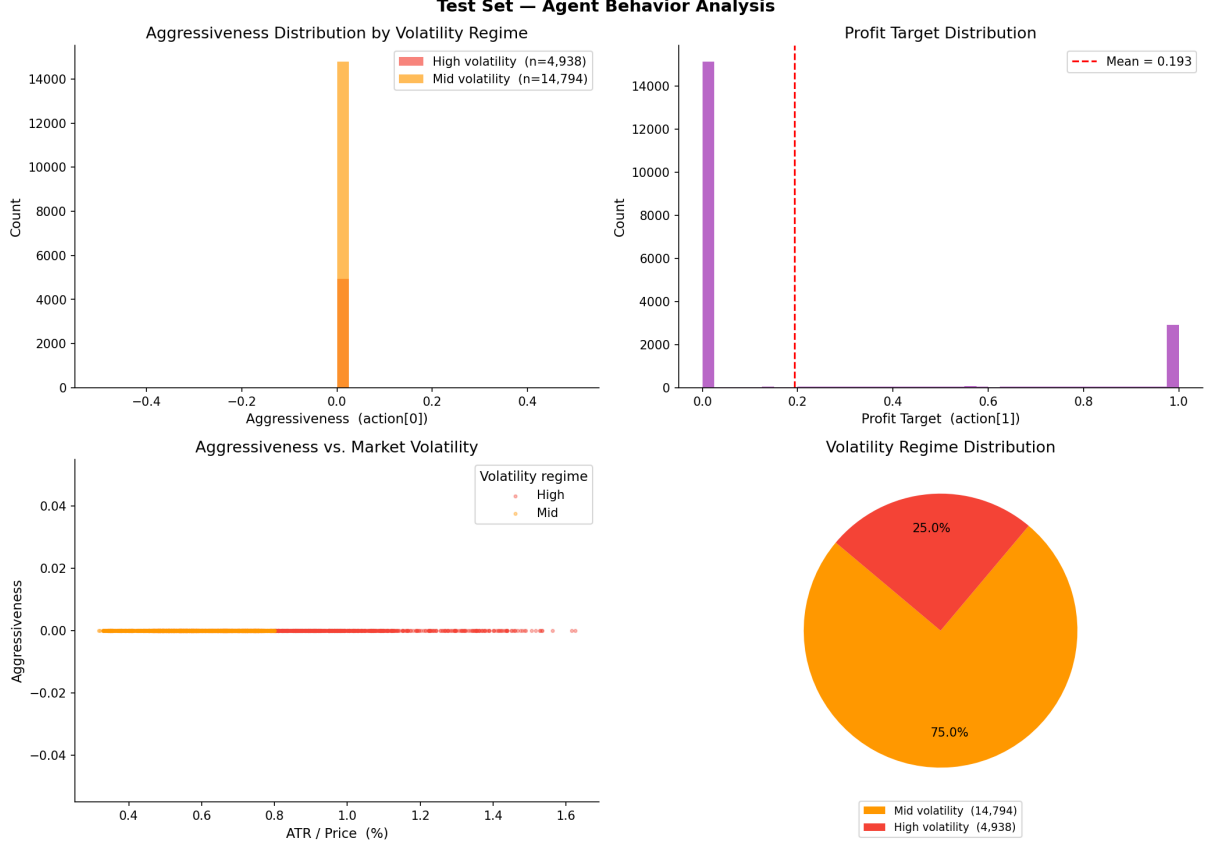


Figure 6: Action distributions on the test set. **Top-left/Bottom-left:** both aggressiveness and profit target are concentrated at exactly 0.0 across all volatility regimes, indicating a saturated constant policy. **Bottom-right:** the regime distribution across the test period.

Contrary to the expectation of RQ2, the trained policy outputs a near-constant action vector $\mathbf{a} \approx (0, 0)$ deterministically, regardless of the observed volatility regime. Inspection of the policy network confirms this: the raw Gaussian mean for aggressiveness is -9.19 (tanh-squashed to ≈ -1 , rescaled to ≈ 0), and for profit target -4.30 (rescaled to ≈ 0). No statistically meaningful difference in action values exists across Low, Mid, and High volatility regimes.

Why did the policy saturate? The root cause lies in a structural coupling between Bayesian coefficient optimization and the RL action space. The order-placement formula is:

$$\Delta_t^{\text{sell-mkt}} = r_t \cdot (A_s + a_{\text{pt}} \cdot B_s). \quad (20)$$

Bayesian search drove A_s to its lower bound (0.101), making the minimum action value ($a_{\text{pt}} = 0$) already optimal: any increase in a_{pt} only widens the sell gap, reducing cycle frequency. The RL agent correctly learned to keep $a_{\text{pt}} = 0$, but this renders state-conditioned adaptation unnecessary. In other words, the Bayesian stage solved the gap-sizing problem at the coefficient level, leaving no residual degrees of freedom for RL to exploit.

Implication. RQ2 cannot be answered with the current design. The result does not imply that volatility-regime adaptation is impossible or unhelpful — it implies that the coefficient optimization removed the mechanism through which adaptation would have been expressed. A

redesigned system in which coefficients fix the *range* of behavior while RL controls the *selection within that range* is the natural next step (Section 7).

6.3 Cycle Behavior

Figure 7 characterizes the 9,520 trading cycles completed during the test period.

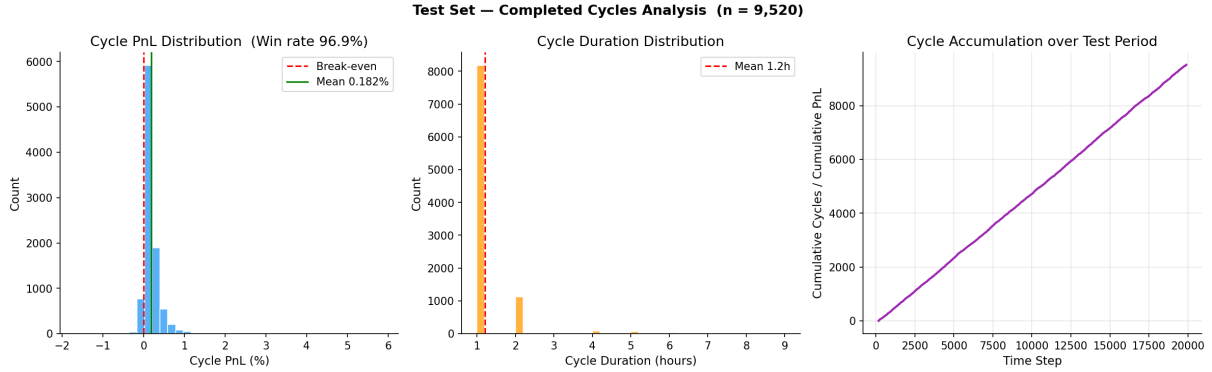


Figure 7: Test-set cycle analysis. **Left:** PnL distribution (win rate 96.9%, mean +0.182%). **Center:** cycle duration distribution (mean 1.2 bars $\approx$ 1.2 hours). **Right:** cumulative compounding of per-cycle returns on a log scale.

The 96.9% cycle win rate and 1.2-hour average duration demonstrate that the agent operates in a near-market-making mode: opening and closing positions rapidly, collecting small spreads on each cycle. The log-linear cumulative return curve (right panel) confirms that the compounding occurs steadily without catastrophic drawdowns, consistent with the 3.12% MDD.

6.4 Limitations

Market impact. The simulation assumes infinite liquidity: limit orders are filled at the stated price with no slippage. At the simulated capital scale (\$10,000), this assumption is reasonable for BTC/USDT (daily volume > \$30B). However, scaling to larger positions would require a market-impact model.

Fee model. A flat 0.05% maker fee is applied. In practice, taker fees (0.10%), funding rates, and spread costs would further erode returns, particularly for the fast-cycle strategy that generates many fills.

Training horizon. The train set (2020–2022) contains only one major bull-bear cycle. Extended coverage of additional market regimes would strengthen the generalization claim.

Policy saturation. The learned policy outputs a near-constant (0,0) action because Bayesian optimization drove A_s to its lower bound, making the minimum action permanently optimal. This is a structural design limitation: co-optimizing formula coefficients and the RL policy for the same objective removes the policy’s degrees of freedom. Future work must decouple these two optimization stages.

A_s boundary. The optimal $A_s = 0.101$ sits at the lower bound of the Bayesian search space, suggesting the Bayesian stage solved the sell-gap problem completely, leaving no room for RL to

contribute further refinement.

7 Conclusion

We presented a non-directional reinforcement learning framework for dynamic grid trading on BTC/USDT in which a PPO agent controls ATR-proportional limit-order placement without directional price prediction. The key contributions are: (1) a five-dimensional MDP formulation encoding both market conditions and portfolio state; (2) an eight-coefficient ATR-scaling formula that keeps order semantics interpretable; and (3) a Bayesian search procedure that improved validation Sharpe by +145% over hand-tuned defaults.

The final model achieves a test-set Sharpe of 43.04 — $29.2\times$ the best fixed-grid baseline — with a 3.12% maximum drawdown. Critically, post-hoc analysis shows the policy converges to a near-constant $(0, 0)$ action output: the Bayesian optimization drove the sell-market base coefficient A_s to its lower bound, making the minimum action permanently optimal and eliminating the need for state-conditioned selection. The performance advantage over fixed baselines therefore stems primarily from the Bayesian-optimized coefficient set rather than from learned adaptive behavior.

This finding reframes the research contribution: rather than demonstrating that RL *adapts* to market regimes, this work demonstrates that (a) the ATR-proportional framework with Bayesian-optimized coefficients constitutes a strong fixed strategy in its own right, and (b) the co-optimization of coefficients and RL policy creates a structural conflict that prevents regime adaptation from emerging.

Next-phase design. To enable genuine regime adaptation, three changes are required:

1. **Decouple coefficient and action roles.** Coefficients should fix the economically meaningful *range* (e.g., sell-market in $[0.1, 1.0] \times r_t$), using Bayesian findings as informed bounds rather than optimized values. RL action then selects *within* that range based on state.
2. **Add directional state features.** Include rolling log-return features (e.g., 1-day and 1-week trends) so the policy has the signal needed to distinguish bull, bear, and sideways regimes.
3. **Monitor policy entropy during training.** Track action standard deviation at each evaluation checkpoint; collapse toward zero is an early warning of the saturation failure mode identified here.

Additional extensions include SAC for improved sample efficiency in the continuous action space, and integration with the Binance live API for paper-trading validation.

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